Wenhao Tang

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Research Interest

My research interest is in programming languages, in particular types and effects. My work focuses on making it simpler to develop reliable software by designing advanced type systems that provide strong static guarantees.

Education

The University of Edinburgh, Edinburgh, UK — Oct 2022 – Dec 2026 (expected)

Ph.D. Student in Informatics
Supervisor: Prof. Sam Lindley

Peking University, Beijing, China — Sept 2018 – July 2022

B.S. in Computer Science and Technology (summa cum laude)
Supervisor: Prof. Zhenjiang Hu

Experience

KU Leuven, Leuven, Belgium — June 2021 – Sept 2021

Research Intern, supervised by Prof. Tom Schrijvers

Publications

Refereed Conference and Journal Papers

POPL **Wenhao Tang**, Shengyi Jiang, Aghilas Y. Boussaa, Sam Lindley, Bruno C. d. S. Oliveira. Bidirectional Typing with Freezing, Skeletons, and Ghosts. *54th ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*, 2027 (to appear). (CORE-A*)

POPL **Wenhao Tang**, Sam Lindley. Rows and Capabilities as Modal Effects. *53rd ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*, 2026. (CORE-A*)

OOPSLA **Wenhao Tang**, Leo White, Stephen Dolan, Daniel Hillerström, Sam Lindley, Anton Lorenzen. Modal Effect Types. *ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2025. (CORE-A)

JFP **Wenhao Tang**, Tom Schrijvers. From High to Low: Simulating Nondeterminism and State with State. *Journal of Functional Programming (JFP)*, Volume 34, e16, 2025. (CORE-A)

LMCS Roger Bosman[†], Birthe van den Berg[†], **Wenhao Tang**[†], Tom Schrijvers. A Calculus for Scoped Effects & Handlers. *Logical Methods in Computer Science (LMCS)*, Volume 20, Issue 4, 2024. (CORE-A)

POPL **Wenhao Tang**, Daniel Hillerström, Sam Lindley, J. Garrett Morris. Soundly Handling Linearity. *51st ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*, 2024. (CORE-A*)
ACM SIGPLAN Distinguished Paper Award

OOPSLA **Wenhao Tang**, Daniel Hillerström, James McKinna, Michel Steuwer, Ornela Dardha, Rongxiao Fu, Sam Lindley. Structural Subtyping as Parametric Polymorphism. *ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2023. (CORE-A)

JFP Hanliang Zhang, **Wenhao Tang**, Ruifeng Xie, Meng Wang, Zhenjiang Hu. Contract Lenses: Reasoning about Bidirectional Programs via Calculation. *Journal of Functional Programming (JFP)*, Volume 33, e10, 2023. (CORE-A)

Refereed Extended Abstracts

HOPE Aghilas Y. Boussaa, **Wenhao Tang**, Sam Lindley. Higher-Order Fork, Modally. *Workshop on Higher-Order Programming with Effects (HOPE)*, 2026.

IMLA Anton Lorenzen, **Wenhao Tang**, Sam Lindley. A Value Trick for Modal Type Systems. *Workshop on Intuitionistic Modal Logic and Applications (IMLA)*, 2026.

ML Family **Wenhao Tang**, Shengyi Jiang, Bruno C. d. S. Oliveira, Sam Lindley. Freezing Bidirectional Typing. *ML Family Workshop*, 2025.

HOPE **Wenhao Tang**, Sam Lindley. Rows and Capabilities as Modal Effects. *Workshop on Higher-Order Programming with Effects (HOPE)*, 2025.

POPL SRC **Wenhao Tang**. Session-Typed Effect Handlers. *Student Research Competition, 51st ACM SIGPLAN Symposium on Principles of Programming Languages*, 2024.

([†] represents equal contributions)

Awards

- Distinguished Paper Award, POPL 2024 — ACM SIGPLAN, 2024
- Third Place in the ACM Student Research Competition (Graduate Category) — ACM, 2024
- School of Informatics PhD Scholarship — The University of Edinburgh, 2022
- John Hopcroft Scholarship — Peking University, 2021
- Award for Academic Excellence — Peking University, 2020

Academic Service

- **Web Co-Chair:** ICFP 2027
- **Program Committee Member:** IFL 2025
- **External Reviewer:** POPL 2027
- **Journal Reviewer:** TOPLAS 2025, JFP 2024
- **Mentor:** SIGPLAN-M 2026

Teaching

- **TA**, *Elements of Programming Languages* — The University of Edinburgh, Autumn 2024
- **TA and Tutor**, *Elements of Programming Languages* — The University of Edinburgh, Autumn 2023
- **TA and Tutor**, *Elements of Programming Languages* — The University of Edinburgh, Autumn 2022
- **TA**, *Introduction to Computer Systems* — Peking University, Autumn 2020

Selected Talks

- *Modal Effect Types*. Seminar, ETH Zurich, Zurich, Switzerland. — Sept 2026
- *Modal Effect Types*. Seminar, University of Southern Denmark, Odense, Denmark. — Sept 2026
- *Ergonomic and Expressive Effect Types*. Seminar, The University of Hong Kong, China. — April 2026
- *Rows and Capabilities as Modal Effects*. POPL 2026, Rennes, France. — Jan 2026
- *Rows and Capabilities as Modal Effects*. HOPE Workshop 2025, Singapore. — Oct 2025
- *Freezing Bidirectional Typing*. ML Family Workshop 2025, Singapore. — Oct 2025
- *Modal Effect Types*. OOPSLA 2025, Singapore. — Oct 2025
- *Modal Effect Types*. EHOP Workshop 2025, Edinburgh, UK. — Aug 2025
- *Effects, Linearity, and Modalities*. Seminar, Peking University, Beijing, China. — Aug 2024
- *Effects, Linearity, and Modalities*. Seminar, Shanghai Jiao Tong University, Shanghai, China. — Aug 2024
- *Soundly Handling Linearity*. POPL 2024, London, UK. — Jan 2024
- *Structural Subtyping as Parametric Polymorphism*. OOPSLA 2023, Cascais, Portugal. — Oct 2023
- *Soundly Handling Linearity*. Seminar, University of Bristol, Bristol, UK. — Oct 2023
- *Soundly Handling Linearity*. EHOP Workshop 2023, Edinburgh, UK. — July 2023
- *Tracking Linear Continuations for Effect Handlers*. Huawei-Edinburgh Joint Lab Workshop, UK. — June 2023
- *Tracking Linear Continuations for Effect Handlers*. SPLS, Heriot-Watt University, Edinburgh, UK. — March 2023